

Scalability & Future Plan

Date	28 Sept 2026
Team ID	6ab221fbe880316a482c6241
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	1 Mark

Scalability & Future Plan: Script-Controlled ACL – Restrict Record Access Based on Field Value

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or expanded features in the future. It also explains the team's roadmap for extending and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Field value and role names are hardcoded in the ACL script.	Any change needs the script to be edited by an admin, which is slow and error-prone.	High
2	The ACL is applied to only one table.	Records in other tables are not protected by the same rule.	Medium
3	Testing is done manually through impersonation.	Changes may break access rules without being noticed.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Works for a small group of users with a few roles.	Use role/group-based access so new users get access by group assignment without script changes.
2	Data Storage	A small number of records on a single table.	Archive old or closed records and index the field used in the ACL condition.
3	Performance	Script is evaluated for every record; acceptable for small lists.	Keep the script lightweight, avoid database queries inside the ACL, and store values in system properties.
4	Security	Access restricted by field value and user role.	Add audit logging, review ACLs periodically, and extend to field-level ACLs.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Move field values and roles into system properties instead of hardcoding them.	1 Month	Easy maintenance without editing the ACL script.
Phase 2	Extend the ACL to other tables and add field-level ACLs.	2–3 Months	Wider protection of sensitive data across the platform.
Phase 3	Add automated tests and audit reporting for access control.	4–6 Months	Reliable, verifiable and compliant access control.